Let $x \sim p_x(x)$ be a random variable and $y = f(x)$ where f is a deterministic function.

0.0.1 Invertible transformations

Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be a bijective (invertible) function. We have

$$p_y(y) = p_x(f^{-1}(y)) \left|\det[\mathbf{J}_{f^{-1}}(y)]\right|$$

where $\mathbf{J}_{f^{-1}}(y)$ is the Jacobian of the inverse function f^{-1} evaluated at y.

0.0.2 Non-invertible transformations

If f is not invertible, we can still compute the distribution of y by considering the pre-image of y under f. Let $f^{-1}(y)$ denote the set of all x such that $f(x) = y$. Then, we have

$$p_y(y) = \sum_{x \in f^{-1}(y)} p_x(x) \left|\det[\mathbf{J}_f(x)]\right|^{-1}$$

0.0.3 Monte Carlo approximation

If f is complex or high-dimensional, we can use Monte Carlo methods to approximate the distribution of y. We can sample $x_i \sim p_x(x)$ and compute $y_i = f(x_i)$. The empirical distribution of y_i will approximate $p_y(y)$. Specifically, we can estimate the density at a point y as

$$\hat{p}_y(y) = \frac{1}{N} \sum_{i=1}^{N} \delta(y - y_i)$$

0.0.4 See also

0.0.5 References